

Maël Madec

Robotics & Autonomous systems Engineer

Engineer in robotics and autonomous systems, focused on the design and characterisation of autonomous systems in real-world conditions. Experience in underwater systems and manipulator arms at Ifremer and real-time multi-sensor monitoring at Centrale Nantes. Python, C++, ROS 2. English proficiency: C1 (TOEIC 980/990).
Available for opportunities across France and Europe.

Contact

+33 6 40 17 67 29
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Open to EU/US relocation
Brest, Brittany, France
LinkedIn profile

Technical Skills

- **Robotics & control:** modeling, ROS2, control systems
- **Mechatronics:** Electronics, industrial automation
- **Embedded Systems:** RTOS, Arduino, STM32
- **Data & Algorithms:** Machine Learning, Fuzzy Logic, sensor fusion, optimization algorithms
- **Programming:** Python, C++, MATLAB
- **Development Tools:** Git, Docker, GitLab CI

Soft Skills

- Adaptability
- Problem-Solving
- Collaborative
- Autonomy
- Teamwork

Languages

Français: Native
Anglais: Fluent (TOEIC 980/990)

Professional Experience

Graduate Engineering Intern — Sep 2025 - Mar 2026

Ifremer - Underwater Systems Unit

- Contributed to the innovation of the Nautile's manipulator arms.
- Prepared a state-of-the-art review of underwater robotics and the Nautile submersible.
- Developed simulation and modelling tools, as well as algorithms in Python, C++ and ROS 2, to characterise the performance of multi-axis robotic arms.
- Designed data-fusion algorithms for arm pose estimation in complex underwater environments.

Assistant Engineer Intern — Sep 2024 - Fev 2025

Ecole Centrale de Nantes, GeM Laboratory

- Designed a real-time monitoring system for metal additive manufacturing, including multi-sensor data acquisition, processing and fusion in Python/C++.
- Modelled and simulated sensors in Python/MATLAB for characterisation, correction and calibration.

Education

Engineering Degree — Sep 2020 - Jun 2026

Bretagne INP - ENIB

- General engineering curriculum covering computer science, electronics, mechanics and control engineering.
- Specialization in Mechatronics, Embedded Systems and Robotics.

Exchange Program in Slovenia — Fev 2025 - Jun 2025

University of Ljubljana - Slovenia

- Robotics projects, including team-based manipulator-arm control and the design of control algorithms.
- Coursework and projects in mechatronics, robotics, control engineering, AI and autonomous systems.

Exchange Program in Italy — Fev 2022 - Jul 2022

Politecnico di Torino - Italy

- English-taught coursework in embedded systems and networks.
- Developed adaptability in a multicultural environment and technical